Md Afzal Hossain

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Education

Industrial and Production Engineering, *(2020 – 2025)*
Shahjalal University of Science and Technology, Sylhet, Bangladesh.
CGPA : 3.44/4.00

Major Courses: Ergonomics, System Modeling & Simulation, Operations Research (Linear, Dynamic & Integer Programming, Decision Analysis), Engineering Statistics, Quality Control and Management, Operations Management, Project Management, Industrial Management, Supply Chain Management, Production System Optimization and Advanced Manufacturing Process, etc.

Higher Secondary Certificate (HSC), Science, *2019*
Sylhet Govt. College, Sylhet, Bangladesh. GPA: 4.92/5.00

Secondary School Certificate (SSC), Science, *2017*
The Aided High School, Sylhet, Bangladesh. GPA: 5.00/5.00

English Proficiency

- **IELTS Academic: Overall Band Score 7.5** [Listening: 8.5, Reading: 8.0, Writing: 6.5, Speaking: 6.0]

Research Interests

⇒ **Operations Research & Manufacturing Optimization:** Formulating bi-objective and multi-objective optimization models for real-world task allocation and capacity planning problems in industrial production lines.
⇒ **Supply Chain Management & CODP Positioning:** Researching Customer Order Decoupling Point (CODP) placement strategies and their empirical effects on supply chain performance outcomes across manufacturing sectors.
⇒ **Manufacturing Systems & Lean Production:** Applying Lean tools, takt-time analysis, and activity sampling to improve production-line efficiency and reduce operational waste.
⇒ **Data Analytics & Simulation for Decision Support:** Using Discrete-Event Simulation (Arena) and Data-Driven Modeling (PLS-SEM) to Support Decision-Making in Production, Service, and Healthcare Operational Systems.

Research & Thesis

- **Undergraduate Thesis:** *Validating the Drivers of Customer Order Decoupling Point (CODP) and Their Impact on Supply Chain Performance: A PLS-SEM Approach in the Furniture Industry* 2025
  Validated CODP drivers using PLS-SEM and identified product customization as a significant determinant, revealing its downstream trade-offs with service level and cost efficiency in furniture manufacturing..

Publications

[IEOM 2025] **Md Afzal Hossain** *et al.* **A Bi-Objective Optimization Model for Real-World Task Allocation in RMG Sewing Lines: Capacity Maximization and Minimization of Capacity Loss**. DOI: 10.46254/BA08.20250363

[IEOM 2025] **Md Afzal Hossain** *et al.* **Determinants of CODP Positioning and Their Effects on Supply Chain Outcomes: An Empirical Investigation in the Furniture Sector**. DOI: 10.46254/BA08.20250373

[IEOM 2025] **Md Afzal Hossain** *et al.* **Data-Driven Prediction and Recommendation of Elective Courses for Master's Students**. DOI: 10.46254/BA08.20250209

Projects

1. **Bakery Factory Takt-Time Reduction**
   Applied Lean manufacturing tools to analyze and reduce takt time in a bakery production line, identifying process bottlenecks and improvement opportunities.
2. **Hospital Diagnostic System Simulation**
   Built a discrete-event simulation of a hospital's diagnostic test system using Arena to evaluate patient flow and resource utilization.
3. **Activity Sampling Study – Furniture Panel Section**
   Conducted an activity sampling study of the edge-banding zone within a panel-processing section to assess workstation utilization and productivity.
4. **Agricultural Supply Chain Resilience Case Study (ISCEA 2024)**
   Developed a case-based strategy for strengthening agricultural supply chains through technology adoption and resilience planning for an industry client.
5. **RC Water Trash Collector**
   Designed a remote-controlled, sustainable, eco-friendly device for aquatic waste collection.
6. **Automated Brightness-Adjustable Bulb**
   Developed an automated brightness-control system as a sustainable solution for street lighting energy efficiency.

Technical and Analytical Skills

- **Data Analysis & Statistics:** SmartPLS, MATLAB, Advanced Excel
- **Simulation:** Arena Simulation Software
- **Engineering Software:** AutoCAD (Autodesk)
- **Office & Presentation Tools:** MS Excel, MS PowerPoint
- **Managerial Skills:** Leadership & Organizing, Production Process Analysis, Presentation, Communication & Teamwork

Achievements

- **Ptak Supply Chain Case Competition 2024** (ISCEA Bangladesh) – Awarded a 60% scholarship (USD 660) after competing against 200+ teams nationwide.
- **IPESphere National Project Fair** – 2nd Runner-Up, competing against participants from 30+ universities.
- **CaseQuest Competition** (SIPEAA) – 1st Runner-Up.

Experience

- **Industrial Attachment / Intern,** *Hatil Complex Ltd.* 1 Month, 2024
  Gained insight into overall factory production processes across manufacturing stages, and practical exposure to production management tools and operational decision-making practices.
- **Industrial Attachment,** *Bangladesh Industrial Technical Assistance Centre (BITAC)* 10 Days, 2024
  Gained hands-on exposure to conventional and non-conventional manufacturing processes through practical industrial training.
- **Industrial Training,** *Khadim Ceramics, Sylhet* 1 Day, 2023
  Observed ceramic manufacturing processes and production-line operations.

Leadership & Activities

- Co-led the Registration & Guest Hospitality team for the 3 Decades of IPE, SUST anniversary celebration, coordinating logistics for a large-scale departmental event.
- Served as a volunteer member at KIN, contributing to community-oriented initiatives.
- Organized and coordinated events for the IPE Department's Sports Week at SUST.

Reference

- **Dr. A. B. M. Abdul Malek**, Professor, Department of Industrial & Production Engineering, SUST, +8801791620777, abmmalek@gmail.com
- **Dr. Abul Mukid Mohammad Mukaddes**, Professor, Department of Industrial & Production Engineering, SUST, +8801777891684, mukaddes-ipe@sust.edu
- **Rafid Ahmed Chowdhury**, Lecturer, Department of Industrial & Production Engineering, SUST, +8801759501929, rafid-ipe@sust.edu